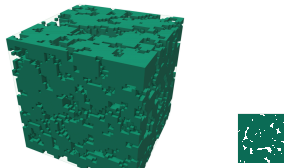


Validation geometries and evidence roles

a–b: real-mask sampled-state transfer; c–g: controlled mechanism tests; h–i: segmented-morphology diagnostics

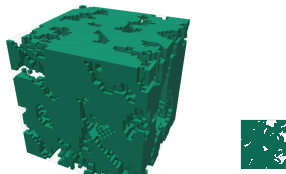
a Bentheimer 225° transfer

real-mask state transfer



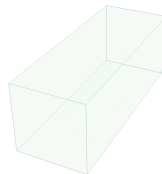
b Public Berea 128° transfer

independent real-mask transfer



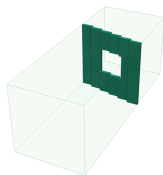
c Orthogonal duct

all-fluid Cartesian limit



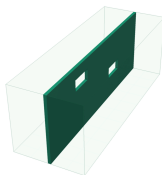
d Skewed duct

non-orthogonal exchange



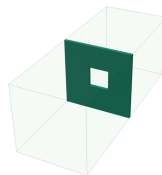
e Thin-wall synthetic

unresolved wall separation



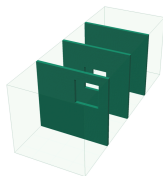
f Narrow-throat synthetic

connected constriction



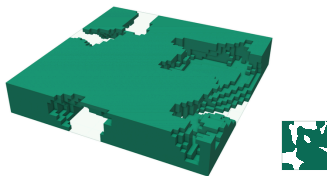
g Maze synthetic

tortuous reachability



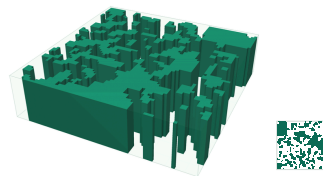
h Bentheimer segmented crop

segmented morphology diagnostic



i Fibrous proxy (SI)

high-wall-contact diagnostic



■ solid / obstruction

□ void / pore space

— computational-domain boundary

■ displayed binary section

inlet → common +x drive outlet